Committee: International Telecommunication Union (ITU)

Topic: The Question of Promoting Future Connectivity through Accelerating 6G Research and Global Spectrum Allocation

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Introduction

In early 2019, the propagation of 5G technology began. This 5G mobile communication was introduced to keep pace with forward-thinking modern technology and build major infrastructure for Industry 4.0. Enhancing frequency bandwidth, 5G opened the door for highly reliable, low-latency wireless communication. 6G, officially named IMT-2030, is currently taking a major step forward today. Though 6G is still in a developing state, chief industries have already begun shifting their focus toward up and coming technology, aiming to set leading technical standards worldwide. Specifically, IMT-2030(6G) is a currently developing 6th generation of mobile technology which is open, secure, and most remarkably high-speed. However, outside of speed acceleration, it is also a future digital infrastructure basis enabling super-realtime communication, ubiquitous connection, and AI or Internet of Things(IoT) combination representatively.

Introduction of IMT-2030 to digital society is not an individual country state's assignment. Reasonably, countries that preempt IMT-2030 technology and standardize it ahead obtain massive economic and military advantages for decades impacting the domestic situation. Acknowledging this pattern, IT powerhouse countries including the United States, China, South Korea, European countries and India are in intense international competition to enact the IMT-2030 standard. Distribution of radio wave resources is one of the key points of this issue. Radio waves are a limited resource of space(or frequencies) that can only be used without overlapping. IMT-2030 utilizes SHF(super-high frequency) bandwidth of 3 to 30 GHz as an universal coverage of the entire network. The rapid worldwide increase in data traffic and the demands of IMT-2030 require an astronomical investment in global digital infrastructure. At the same time, international efforts are essential to prevent technological fragmentation between countries. Furthermore, IMT-2030 development intensifies the digital divide as well. Low population or developing regions such as LEDCs and rural areas lack access to advanced technology and its benefits. On a larger scale, a real-time connection system

connotes a perilous possibility. From mass personal information to national infrastructure management, a single data spill or cyber attacks would be highly likely to paralyze a certain country's whole network and function. And the international community cannot neglect the possibility of cyber war either.

In 2023, ITU already established IMT-2030's future vision through 'Framework and overall objectives of the future development of IMT for 2030 and beyond'. It implies various application and harmonization goals of IMT-2030 while following technical trends. In 2026, ITU presented 'Technical requirements for the 6G future'. Performance requirements, updated usage scenarios, and future resilience, etc., were written reinforcing the IMT-2030 basis. Accordingly, as the usage of 6G became no longer negligible, it's time for the international communities to utilize the technology, while tackling the challenges it poses.

Definition of Key Terms

IMT

The term IMT stands for International Mobile Telecommunications. It is a generic term used by the ITU community to designate broadband mobile systems. It encompasses IMT-2000, IMT-Advanced, IMT-2020(5G) and IMT-2030 collectively.

The development timeline for IMT technologies typically holds a 10-year cycle. This includes future trend investigation, technical and international requirements and final radio interface specifications. In brief history, IMT-2000(3G) is the start of IMT flow. It established a global standard for wireless networks in order to join different and incompatible technologies together. IMT-Advanced(4G) is a faster version of the previous method. Major change contains all-IP networks, massive data forwarding, and speed. Next, IMT-2020 targeted not only fast data of up to 20 Gbps but far-ranging capability of smart city, car and machine connection. In its current state, ITU is already directing 6G's upcoming project to further blend physical society with digital space.

IMT-2030

IMT-2030 is the official framework and name designated for 6G(sixth generation of mobile network and technology). It builds on existing standards for 5G(IMT-2020) networks with enhanced and envisaged capabilities such as speed or low latency to support advanced and immersive utilization.

ISAC (Integrated Sensing and Communication)

ISAC is an innovative technology that allows wireless networks to simultaneously transmit and interchange data to and from user devices and sensed environments. It is also identified as one of ITU’s IMT-2030 major usage scenarios. By analyzing radio frequency(RF) signal reflections, the system integrates information regarding the surrounding environment or object. Moving beyond conventional data transmission, 6G changes the radio channel itself into an environmental sensor. This capability enables effective synergy with artificial intelligence(AI) and autonomous driving and accelerates 6G research and long-term objectives.

Radio Wave

A radio wave is an electromagnetic wave that is often used in communication such as broadcasting, mobile phones, and wi-fi. Its wavelength ranges between 1 millimeter and 30,000 meters and has a frequency between 10 kilohertz and 300,000 megahertz. Specifically, a maximum 300,000 megahertz means vibrating 300 billion times a second, channeling high-capacity data at high speed. This advantageous feature is why countries experience intense international competition to secure stable bandwidth.

However, physical obstacles and dead zones limit their bandwidth. Radio waves travel in a straight line and when they hit large and solid obstacles, they are easily absorbed or reflected. It hinders wave transport creating ‘shadow zones’ where signal is weak or drops out completely. Furthermore, other electronic devices operating on the same frequency rate are also common sources of interference. This vulnerability of radio waves is why frequency allocation is such a sensitive problem.

Bandwidth

Bandwidth is the measured amount of data that can be transferred from one point to another through a network connection in a set amount of time. It mainly concerns frequency capacity and determines how much data a certain network can upload or download at the exact same time. Moreover, it supports high-quality data exchanges, maintains website stability or prevents bottlenecks. While it is closely related to spectrum, the two concepts are clearly different. Spectrum determines which frequency band to use like an address, and bandwidth settles how widely the frequency space is being used. Typically, bandwidth is expressed using bitrate and measurement of bits per second(bps). IMT-2030 internet bandwidth and peak data rates are predicted to accomplish up to 1 terabit per second(Tbps).

Centric-Band Expansion

Centric bandwidth expansion is a new spectrum consideration being discussed worldwide to overcome existing 5G’s limitation and accommodate 6G’s large capacity. A new frequency range is about 7-24 GHz with a special focus on 7-15 GHz, being supported by GSMA(Global System for Mobile communications Association). Discussions with the mobile community continue,

accompanied by proposals for the ITU 2023-2027 World Radiocommunication Conference(WRC) study cycle. (2023-2027 WRC study cycle refers to ITU's four-year study period on the rules governing the uses of radio-frequency and non-geostationary-satellite orbits).

Internet of Things(IoT)

Internet of Things or IoT is a network of collaborations and communications between people and things through physical devices, vehicles, appliances along with software systems. IOT depends strongly on global standards and operates smoothly when it follows them. The potential application is vast and infinite as technologists envision entire 'smart cities' consisting of IoT technologies.

In industrial terms, the IoT method guarantees various advantages and enhancements in modern industry. IoT processes reduce manual repetitive tasks, which optimizes consumption and help businesses improve profitability. Industrial IoT devices are commonly used in manufacturing to supervise and monitor machine performance, detect equipment failures and coordinate production processes. Moreover, the retail industry can utilize IoT devices to track customer behavior, optimize inventory levels and store layouts. Leveraging IoT in supply chains, shipments, and inventory management helps companies significantly boost productivity. On a personal level, smart homes such as home IoT related automation and security devices or smart cities including IoT sensor-based traffic management and digitised healthcare are promising areas for future commercialization. However, increasing security and privacy risks, data overload and high costs also act as hindrances to IoT technology.

Remarkably, IMT-2030 integration plays an innovative role in improving IoT applications' efficiency and range. 6G technology's constant development will conclusively lead to latency reduction, higher data throughput and new learning techniques that benefit IoT's capabilities to cooperate both with land based and satellite based mechanical systems.

Non-terrestrial Networks(NTN)

Nonterrestrial Networks(NTNs) are wireless communication systems that operate above Earth's surface. They are not solely dependent on Earth's terrestrial infrastructure for connectivity and include platforms such as satellites in low Earth orbit (LEO), medium Earth orbit (MEO), and geostationary orbit (GEO). They play a crucial role in enhancing global communication coverage and

fill the gaps left by traditional ground-based telecommunication systems especially in remote or underserved areas.

Digital Divide

The digital divide is the multifaceted gap including accessibility, affordability, quality and relevance between different socio-economic and demographic groups who have quick access to internet or digital technology and those who do not. While developed groups seek new technological advancement, some still struggle to keep pace. Developing harmonised spatial connectivity and providing evidence-based policies is a key way to alleviate digital divide between people and promote world-wide achievement.

Background Information

The ITU IMT-2030 Vision and Current Progress

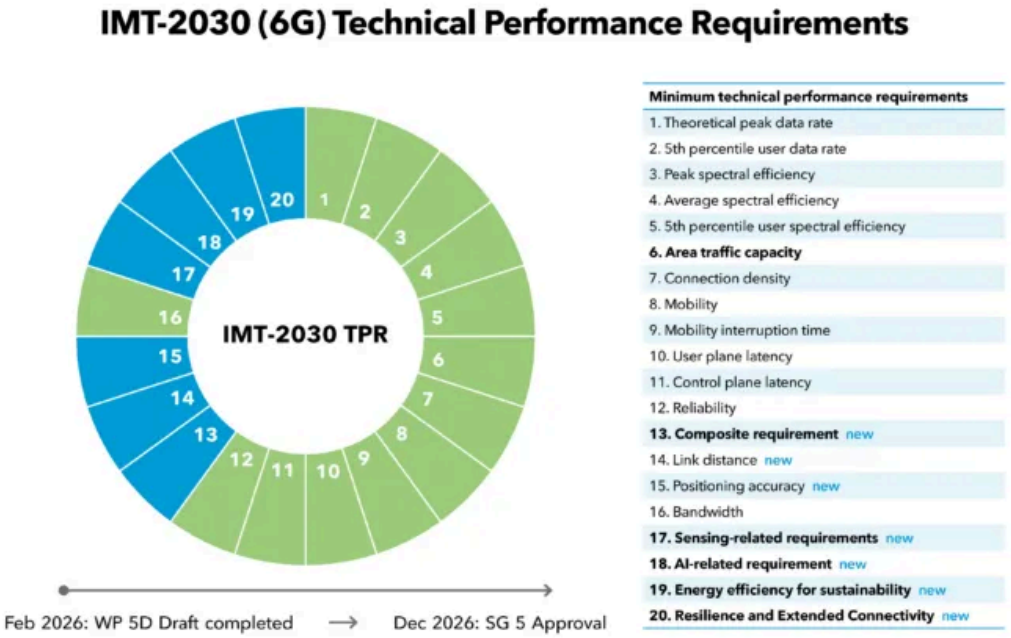
Through the evolution of information and communication technologies, ITU built up its vision and basic directions toward IMT-2030 in 2023 from the document, 'Framework and overall objectives of the future development of IMT for 2030 and beyond'. The motivation for the development of IMT-2030 is to create an inclusive information society toward United Nations Sustainable Development Goals(SDGs) contribution. Going further, ITU marked a major step in 2026 with specified documents.

Part of ITU Radiocommunication Sector(ITU-R), Working Party 5D has defined technical requirements of IMT-2030 through "Minimum requirements related to technical performance for IMT-2030 radio interface". First, ITU-R Working Party 5D is a group responsible for the development and harmonization of IMT(International Mobile Telecommunication) systems. This document outlines 20 core technical performance requirements. These can be grouped into previous IMT-2020 fundamentals, evolution state and newly introduced abilities.

The following are part of present technical performance requirements for IMT-2030(6G) proposed by ITU. Theoretical peak data rate(1) stipulates maximum throughput per device under ideal conditions, targeting 50-200 Gbps, well above IMT-2020's 20 Gbps. 5th percentile user data(2) is the minimum speed experienced by the poorest 5% of users. This term sets a threshold value to ensure fair internet service distribution. It is defined for the purpose of evaluation in the IMT-2030 usage scenario, eMBB. The tested environments are indoor hotspot, dense urban and rural regarding downlink and uplinks. Peak spectral efficiency(3) is the maximum data rate that a communication system can deliver per unit of frequency bandwidth under ideal conditions of no signal disturbance along with best reception conditions. Crucially, it maximizes the efficiency of limited frequency resources through dense, high-speed data delivery. This is achieved by integrating advanced communication technologies such as Massive MIMO (large antenna arrays) and high-order data encoding (M-QAM). Area traffic capacity(6) functions as a basic indicator providing hyper-connectivity including smart homes, wearable devices and immersive services. It suggests that IMT-2030 will widen area traffic capacity beyond IMT-2020.. Connection density(7) represents the maximum number of devices that a network can maintain simultaneously within a certain unit of area

while retaining an acceptable level of service quality. It further promotes IMT-2030's link to IoT(Internet of Things) reaching 1 million to 100 million devices per square kilometer.

In addition to established clauses, seven new capabilities were added as part of IMT-2030 TPR for IMT-2030 standardization. Composite requirement(13) requires not only the primary factor but a composite of various technological qualities mostly TPR aiming to achieve composite future goals. Specifically, it representatively counters trade-off problems which is an increase in latency or reduction of delivery efficiency due to excessive speed rise in wireless connection. Moreover, as ITU-R M.2160 emphasizes performance and sustainability's integration, the composite requirement acts as a key value for 6G shaping. Sensing-related requirements(17) are also a novel concept supporting ISAC(Integrated Sensing and Communication) illustrating standards toward radars sensing signals such as location, direction, speed and shape. This puts extra usability and details to 6G development and connects real-time interaction between objects or infrastructures and the internet. AI related requirement(18) is an undeniable option to further collaborate with AI(Artificial Intelligence) in the present era. It is a first technical performance specification that seeks AI's amalgamation with internal network infrastructure and radio interface. Linked to the usage scenario of AIAC(Artificial intelligence and communication), it functions as a whole AI engine that has learning, inference and compute communication metric advantages.



IMT-2030(6G) Technical Performance Requirements (International Telecommunication Union, 2026)

There are six proposed usage scenarios that IMT-2030 growth visualize. Exploiting TPR, ITU proposes Immersive communication(IC), Hyper reliable and low-latency communication(HRLLC), Massive communication(MC), Ubiquitous Connectivity(UC), Artificial intelligence(AI) and communication (AIAC), and Integrated sensing and communication(ISAC). Within the commercialization settings of IMT-2030, the recommended framework(ITU-R M.2160) defines overarching principles to promote global harmonized foundation. Sustainability, security and resilience, connecting the unconnected and ubiquitous intelligence is not only technically mandatory value but to pursue sustainable and global-standardized technologies in chord.

Apart from excessive advancement of 6G, the global standardization act is very crucial in order to have world-wide bandwidth collaboratively away from fragmentation. Minimizing the gap between followers or recipients and tech leaders will further boost tech evolution. Giving 5G's current distribution as a specific example, 5G coverage has expanded to cover 55% of the world's population in 2025. However, it currently reaches 84% of people in high-income countries, but only 4% of people in low-income countries where 4G remains a reliable connectivity solution. Though relatively high allocation rate, connectivity efficiency differs region by region. Especially, for 6G, global standard curtails significant damages. This enables global roaming of digital devices across borders, financial advantage through lower production cost and overwhelmingly rapid adoption of equal components and interactive connection with society including autonomous vehicles, Urban Air Mobility(UAM), and satellite communication. Moreover, standard setting through a structured, multi-year collaboration led by ITU will ensure clear inclusion of usage scenarios, key performance indicators and equivalent dispensation.

Technological Evolution from 5G to 6G

In tandem with current 6G research efforts, the rollout and adoption of 5G services continue to rise around the world. Like this, 5G and IMT-2030 are relatively linked and contribute to each other's enhancement. 6G networks are being built on the foundation of existing 5G technology and 5G-A infrastructure and further to provide additional capability and reliability improvements. To illustrate, present 5G is an urban-focused network with 20 Gbps of speed, 1ms of latency, and sub-6 GHz, mmWave. On the other hand, 6G is a global based internet of 1 Tbps of speed(theoretical amount yet), 1 microsecond latency, terahertz bands spectrum. It also features AI integration as a core component and ISAC as a unique capability. This clearly demonstrates that 6G's new functionalities will expand the possibilities of the network platform by gradually complementing and eventually surpassing 5G.

Several countries are involved in the field of 6G research and implication in global state through both national and regional research institutions. In Europe, several 6G research projects are leveraging cooperation between public and private parties to advance architecture needed to facilitate and prepare the next generation of wireless connectivity. The major initiative Hexa-X and its successor Hexa-X II constitute IMT-2030 research and the Germany Open 6G Hub functions as a research hub. Hexa-X is the representative flagship national research project for 6G held by the European Union(EU). It focused on establishing fundamental concepts, visions and structure of 6G. Following that, Hexa-X II discusses the way to implement and adopt a 6G network. In the U.S., Next G Alliance promotes successful processes in North America and concentrates on topics including applications, societal and economic needs, sustainability, and technology. China's IMT-2030 Promotion Group established by the Ministry of Industry and Information Technology(MIIT) works for the country's 6G R&D(Research and Development) efforts. In addition, in March 2023, the Ministry of Internal Affairs and Communications (MIC) in Japan earmarked JPY 66.2 billion(about 418.5 million USD) to support the Innovative Information and Communications Technology Fund Project.

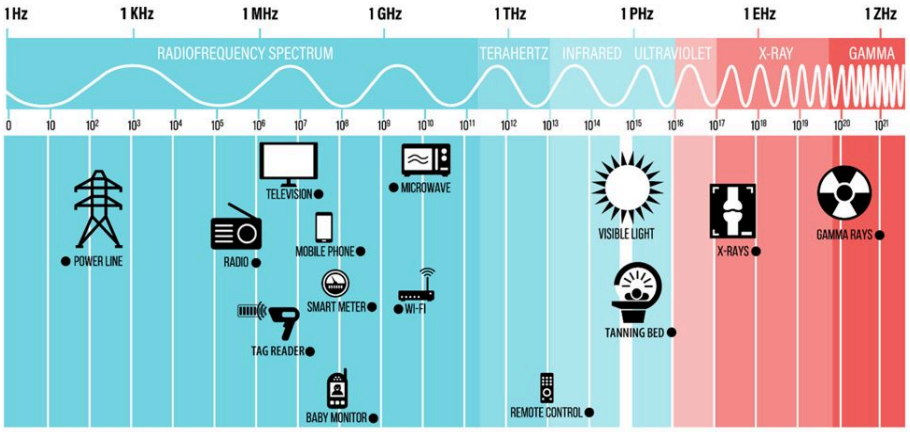
Radio Frequency Spectrum Allocation Dynamics

Spectrum refers to the complete range of electromagnetic frequencies used for communications across a variety of services and industries. It has significant value in economic terms including production of telecommunication services. But, spectrum shortfalls are predicted due to finite amounts and National Regulatory Authorities(NRAs) are setting targets to secure extra spectrum to meet future needs. Depending on how the spectrum is allocated, the characteristic and technological direction of the radio wave used for 6G alters completely. Therefore, establishment and updating of international regulation on the use of radio-frequency spectrum is required firstly. Radio Regulations(RR) is the context adopted by the World Radiocommunication Conference which determines how radio frequency spectrum is shared covering terrestrial fixed and mobile radio services, satellite-based services and more. It clarifies rudimentary standards for radio spectrum to ensure successful coexistence of multi-complex services. World Radiocommunication Conferences (WRCs) review the way specific portions of the radio spectrum are allocated and additionally negotiate and bring on relevant modifications to RR within Member States.

The spectrum distribution's reason is on physical and economical limits to prevent radio waves' collision chaos and share limited resources effectively. Though radio waves are unseeable, they cause radio wave interference or electromagnetic interference(EMI) which is unwanted electrical signal energy disrupting the clear reception of useful radio or wireless signals. In addition, as

mentioned, radio wave's function completely differs depending on the spectrum's frequency. Spectrum is generally divided into three ranges of low, mid, and high frequencies.

Low-band frequencies travel the longest range of 10km with under 1GHz of frequency and covers rural areas and wide geographic coverages passing through walls and buildings. Mid-band frequencies have a medium range having between 1GHz and 10 GHz frequency rate. It is a major band for 5G networks and high-speed broadband which carries more data than low-band. High-band frequencies hold above 10GHz having the shortest distance. As they cannot go through walls or buildings, it can contain areas such as single buildings or stadiums. While 6G aims to operate in high-band frequencies, mixed usages between mid-band and high are most remarkable.



Multiple usages of spectrum (Government of Canada, 2024)

Threshold of IMT-2030

While technical performance requirements function as IMT-2030's guideline, international society also faces challenges of IMT-2030. Navigating the performance thresholds and addressing it would further promote successful implementation of IMT-2030. These are major limitations of 6G. Spectrum allocations are one of the primary challenges for 6G availability. As mentioned above, sufficient spectrum resource distribution and effective interference management systems still lack and put strains between international communities. Terahertz(THz) spectrum, especially, is a pivotal component of 6G to support high data rates and low latency. However, THz waves are easily disrupted by water vapor and oxygen in the atmosphere, limiting their transmission range. Technical limitations include efficient antenna development, signal processing capability, and network components that handle ultra-high data rates and ultra-low latency. These technical goals surpass current hardware and

signal management techniques, so additional research and investments are necessary as a result. Moreover, cost and infrastructure deployment are regarded as crucial challenges too. 6G network development and deployment call for substantial amounts of investment including infrastructure, technology, equipment and research. Therefore, overcoming cost barriers, ensuring affordability for end-users and narrowing the digital divide between countries should be an essential assignment for 6G growth.

Data Breach and Cyber Security Issues

While IMT-2030 offers a wide variety of afresh capabilities and application, there remain risks including massive data breach and cybersecurity perils. With the escalation in connected devices and IoT expansion, more personal and enterprise data are more likely to be exposed than before. Organizations and governments will have bigger access to substantial amounts of personal information concerning ethical and legal issues about privacy violations. Moreover, a broader range of networks enables hackers to exploit weak points to intercept, alter or steal sensitive information. Identity theft and deepfake threats utilizing private information and artificial intelligence are also rising. In case that cyber abuse grows into a country or global scale, the possibility of cyber war might be inescapable. Therefore, while promoting advancement of 6G, countries should enforce strong security systems and harmonious usage.

Digital Divide and Current Situation

IMT-2030's constant advancement paves the way for further technological improvements for the country itself. Currently, 5G as the latest form of network system, reaches 84% of people in high-income countries, but only 4% of people in low-income countries. It indicates that the technological growth has been uneven and there remains high plausibility for IMT-2030 implement as well. In a regional term, Europe covers 74% of population with 5G, Asia-Pacific region with 70%, and the Americas with 60% as well. On the other hand, a considerably low number of 13% of 5G coverage exists in the Arab States region, Africa holding 12% and CIS(Commonwealth of Independent States) with 8% of the population.

Since 3G and 4G technology is not even in a complete form, this disparity needs to be managed alongside IMT-2030 expansion. Sustainable Development Goal 9 (SDG 9) which aims to build resilient infrastructure, promote inclusive and sustainable industrialization and foster innovation with universal access should be highly regarded and pursued to counter these concerns.

Severe digital divides can trigger negative global socio-economic impacts. When these gaps remain unaddressed, they lead to significant disparities particularly between urban and rural areas which are largely driven by infrastructure limitations. And disadvantaged groups including women and young people are more prone to experiencing digital skill divisions. Low levels of digital technology training drive to fewer employment and career opportunities and hindrance to economic growth. A 10% rise in mobile broadband penetration can increase per capita GDP by 1.5-1.6%. Additionally, digital inequality carries major political implications in lower civic engagement or disruption in democratic processes. This clearly demonstrates not only lack of technical development but also social, economic and political aspects.

Possible solutions

Multiple international unions and global conferences continuously contribute effort to 6G development. As 6G is still at a rudimentary stage, there still remain concerns and huge barriers of both technical and physical. These include limited provision of spectrum containing high possibility of countries' dispute and digital divide. Consequently, international communities and countries should cooperate to investigate sustainable steps of IMT-2030 advancement.

Promoting Dynamic Spectrum Sharing

Mutable spectrum sharing systems are especially mandatory for IMT-2030 technology progression. It often refers to DSS(Dynamic Spectrum Sharing). It lets cellular networks run at the same time on the exact same frequency band. Rather than using permanent frequency splits, intelligent base station software dynamically reallocates resources every millisecond based on real-time user demand. This counters the worldwide frequency limit and enables CapEx reduction, which includes early base station establishment and securing of 6G coverage. While it is already utilized in 4G LTE(Fourth Generation Long-Term Evolution) and 5G, it is also a core technology that enables flexible 6G frequency distribution.

Member states may put present technology as a fundamental basis and manage frequency distribution with advanced versions called Multi-Radio Access Technology Spectrum Sharing(MRSS) for 6G. Additional AI management is one of highly anticipated abilities. AI models can function to optimize real-time spectrum scheduling and stop inter-system interference automatically, improving the efficiency within high frequency.

Promoting Integration of Terrestrial and Non-Terrestrial Networks

One current problem is that constructing infrastructure solely with terrestrial networks is both intricate and expensive within low-income countries or remote regions. Implementing NTN(Non-Terrestrial Networks) including satellites, high-altitude platform stations(HAPS) and unmanned aerial vehicles(UAVs) provides wide global coverage and seamless integration without high financial pressure.

However, orbital slots and satellite communication frequencies are easily dominated by high-income countries or major companies firsthand. Therefore to ensure equal access to space 6G network and enable non-terrestrial connection, international communities have needed to reorganize ITU regulatory framework and funding in regards to infrastructure establishment support.

Capacity Building and Financial Support to Diminish Digital Divide

In order to promote world-wide and equitable network development, financial support is an undeniable option. Especially, high-income countries can provide extra infrastructure funds for 6G settlement to low-income counties. These include physical infrastructure such as ground stations, Open RAN(Open Radio Access Network) and technical training centers. Moreover, provision of technical education and training opportunities will ensure citizens to acknowledge the importance of integrated network systems.

In a community level solution, open RAN(Radio Access Network) could build up flexible and approachable networks using standardized open interfaces. This has an advantage of expense reduction and economic feasibility that lightens up the burden of economically developing countries. Member states must provide open architecture based small cells for local groups to independently establish and operate low-cost base stations while reducing their dependency on major telecom companies.

Implementing Green 6G and Energy-Efficient Standardization of IMT-2030

While Base Transceiver Station(BTS) and billions of digital devices execute hyper-connectivity society, its immersive emission of carbon is also not negligible to promote sustainable technology. Energy efficiency standard and reduction of carbon footprint are to be included as key performance indicators in IMT-2030 standardization framework. Utilizing this global framework, transfer of green telecommunication technologies to developing countries will further mitigate environmental damage while also reducing digital divide gaps. Fostering Eco-Friendly technologies such as Zero-Energy Devices(ZEDs) or renewable energy base stations will enhance the energy efficiency of IMT-2030 as well.

Reinforcing Cybersecurity

Cybersecurity threats and cybercrime concerns are continuously being raised. To mitigate those in hyper connected 6G and IoT environments, international governing communities should offer and mandate a ‘Security by Design' framework for most infrastructure builds. Security by Design refers to a proactive cybersecurity method which enhances security level into software, hardware and digital systems. They should include secure defaults such as MFA(Multi-Factor Authentication), defense in depth and separation of duties.

Moreover, member states are highly encouraged to establish cross-border threat information platforms in order to enable real-time danger detection and neutralization of cyber-attacks worldwide. Before possible dangers impair critical national infrastructure or even on a global scale, highly intellectual bodies and technological centers of each member state could collaborate for a further harmonious 6G society.

Promoting Digital Literacy and Skill Development in Marginalized Areas

Digital literacy is the cognitive and technical capability to use digital technology such as communication tools and networks safely and effectively. Lack of digital literacy and basic skills is a fundamental matter of digital divide between countries or stratums. Gaps do not decrease when enough physical network and equipment is provided but education and skill development are deficient between people. Therefore, this must be accompanied by education, particularly targeting developing nations, rural communities, and underprivileged populations.

Establishing regional digital training academies or programs and curricula through cooperation is required. Moreover, focus on marginalized groups and rural residents can boost the efficiency overall. Financial and instrumental support including safety guidelines, technological advisors, and education materials from developed countries and international organizations could contribute to global digital divide alleviation too. This initiative will ensure that wide-range citizens acquire the essential and basic skills needed to safely navigate and leverage fast changing 6G societies.

Major parties involved

Japan

Japan's IMT-2030 development state and pace is in the highest class globally and leading group. It remains as a key force in the next generation of telecommunication and aims to introduce IMT-2030 mobile data technology by 2030 to sustain its leadership in digital market and innovation. Japan's key focus on global 6G research and development is holographic and AR(Augmented Reality) and VR(Virtual Reality). Holographic usually refers to a hologram, a three-dimensional image created using laser light, interference and diffraction. Japan has advanced these technologies from an early period to form an interactive environment. In conclusion, Japan is positioning itself as a global leader in 6G and forwards to develop through aggressive R&D(Research and Development) trials and international partnership.

The Japanese government also has committed a substantial amount of resources and financial investments for 6G development by pledging billions of yen. The government also expects to commission works or projects to private companies and universities to further promote the fast progress. Remarkably, Japan's leading 6G companies are NTT Docomo and Sony who focus on AI-native networks, sub-terahertz frequency propagation and energy efficient integrations.

Beyond 5G Promotion Consortium is a joint public-private organization launched in December 2020 by Japan's Ministry of Internal Affairs and Communications(MIC). It is considered as a major and anticipatory group comprised of major corporations, universities and global collaborators including NTT and NTT Docomo and plays a central role in research coordination, pilot trials and international standard cooperation overall. In addition to this, MIC continues to expand Beyond 5G(6G) Fund Project and Japan U.S. joint R&D initiative in order to seek for global cooperation within competitiveness.

People's Republic of China

Along with progressive stance of Asian countries such as South Korea or Japan, China is currently building the basis and system for IMT-2030 as well. It has already made headlines by launching an experimental satellite to test THz signal transmission and aims to launch 6G as commercial services before 2030 as a whole. Since IMT-2030's terrestrial and non-terrestrial

combination is one of core capabilities, this achievement is a significant step toward 6G. CTC(China Telecom Corporation) has already received a national invention patent for 6G satellite-terrestrial integration technology. Leading companies include giant tech industries of Huawei, ZTE establishing infrastructure and device manufacturing processes for future wireless technology.

While it stands in contradiction toward the US technology blockade, China aims to adopt an international approach toward telecommunication and even facilitate IoT based intelligent entities like robots and metaverse systems. China is especially focused on international exchange between regions like the European Union, South Korea and India.

United States of America

Since the United States has been at the forefront of technological innovation in longterm, IMT-2030 implication is not an exception to them too. Key goals of the U.S. could be defined as AI-integrated 6G following the current AI trend. Additionally, USA propels early global leadership in 6G commercialization by 2029 to 2030 as its representative position. Operating under the U.S. Executive Branch, the National Telecommunications and Information Administration (NTIA) serves as the core regulator managing federal spectrum usage, broadband expansion grants, and public safety support. Entity corporations mainly Qualcomm, AT&T, and the Next G Alliance are also contributing to shape the 6G roadmap for global improvement.

According to Whitehouse Presidential Action, spectrum relocation presents as necessary actions. It urges the Assistant Secretary of Commerce for communication and Information and more to immediately commence studies with 7.125-7.4 GHz spectrum frequency operations. Spectrum designation similarly requires identifying the band of spectrum frequencies for reallocation for full power commercial licensed use cases.

The European Union

In a perspective of IMT-2030 growth, the European Union works collaboratively as a group rather than individually to avoid the mistake of 5G implementation latency. The EU set sights on achieving technological sovereignty and global leadership in 6G with active commercial deployments and international standard shaping. During the 5G era, European countries had experienced losing competitiveness due to dispersed markets and late introduction. Now, the EU has several large-scale initiatives such as the Horizon Europe Program, the key funding program for research and innovation.

While underlining sustainable 6G networks, it fosters partnerships between academia, industry and government to occupy standards for 6G. In addition to this, the EU has launched a 104 million euro initiative and the latest call marks the first funding round of 2025 and ongoing multi-year commitment to the next generation 6G network.

Republic of Korea

Following its successful roll out of 5G adoption and enabling fast connectivity for the first time in the world, South Korea is now a contender in the 6G race. It aims to secure early global leadership position in 6G technology as well as other competitive countries by integrating next-generation mobile works within the broader term of artificial intelligence and digital infrastructure. In late 2023, the Ministry for Science and ICT(MSIT) announced USD 325 million investment to develop “pre-6G” technology along with goals of being the first country in the world to deploy commercial 6G. It further adds up to the previously established K-Network 2030 strategy to secure 6G technology, promote software-based network innovation, and strengthen the supply chain.

South Korea’s aspiration is highly regarded and expectable looking through their 5G implementation journey. To simply summarize, 4G service was initialized in 2011, 5G standardization organization in 2012, 5G R&D started in 2013, ITU reflected Korea’s strong technologies officially. In 2018, they suggested a 5G candidate technology, and in 2019 they successfully rolled out 5G for the first time in the world. From moderate and weak technical skill levels in the 1950s and 1960s, South Korea now stands as one of the most well-digitized Asian countries.

Federal Republic of Nigeria

While previously demonstrated countries are fast developers in digital infrastructure, there are still lagging countries facing the digital divide. Nigeria is one of those which faces barriers to the adoption of technology. The root causes of this status are power deficits, connectivity and fiber gaps and economic hurdles. Nigeria views IMT-2030 as a future priority while balancing the immediate need to expand basic 4G and 5G. To operate advanced connectivity such as IMT-2030, it costs a high operating cost with expensive diesel generators and service expenses.Not only the government but also rural residents struggle with these costs, as residents cannot easily afford smartphones or high quality data plants lowering investment toward these policies. In a technological aspect, the unreliable grid of Nigeria that fails to provide continuous power hinders fluent link of networks. Moreover, about 44.6% of Nigerians, largely in the North are either without basic telephony service or scarcely

served. And fiber optic cables, a high-speed data transmission medium made of ultra thin strands of pure glass or plastic, are largely concentrated in urban areas like Lagos and Abuja leaving rural areas disconnected.

Confronting such problems, the Nigerian government is constantly putting efforts to establish digital technologies as well as 6G. In December 2023, Nigeria signed a global Act for adoption of IMT-2030 and other facilities related to radio and communications. In addition, Dr Bosun Tijani, minister of Communications, Innovations and Digital Economy signed the Act on behalf of Nigeria at the concluded WRC 2023. These acts clearly indicate Nigeria's positive attempt toward IMT-2030. To actualize, not only the independent country itself but also global attention and cooperation is required to overcome digital and economical barriers.

Republic of Yemen

Yemen is another country deficient in digital connectivity across regions. It does not have a formal national position or policy regarding 6G technology yet since its telecommunication system still remains focused on basic infrastructure maintenance and baseline 4G expansion. Political division and blockades are a destructive obstacle that Yemen is facing in its current state. More than a decade of civil war have destroyed their physical networks, blocked technology imports and split the telecommunication sector within split governance. Continuous infrastructure sabotage, import restrictions and power crisis have delayed modernization, making it hard to maintain stable 3G or basic 4G services. Moreover, high costs of IMT-2030 establishment and government's low investment and attention are highlighted grounds as well.

Timeline Of Events

Date	Description of Event
Establishment of WRC-19 Agenda for High Frequencies, 2019, November	WRC(conference of 2019, ITU) concluded with decisions to study spectrum frequencies above 275-450 GHz and established the early regulatory basis for sub-THz 6G research.
Approval of the IMT-2030 Framework, 2023, November	ITU-R officially adopted Recommendation ITU-R M.2160 to define global vision including six key usage scenarios and primary design principles for IMT-2030.
Adoption of WRC-27 Agenda Item 1.13, 2023, December	At WRC-23, member states researched and debated the way to enable coexistence of satellite and land based frequency. It highlights NTN(Non-Terrestrial Network) and Direct to Cell technology and aiming to discuss these materials in WRC-27.
Initiation of Technical Performance Requirements by ITU-R WP 5D, 2024, June	ITU-R Working Party(WP) 5D began to formulate minimum technical performance requirement reports and evaluation standards for IMT-2030 radio interfaces.
Completion of initial 6G Channel Modeling Frameworks, 2025, November	ITU-R WP 5D completed introductory radio propagation models and electromagnetic compatibility frameworks for 6G frequency bands.
Regional Inter-Organization Coordination Meetings, 2026, June	Regional telecommunication bodies such as APT, CEPT, CITEL, ATU gathered sharing initial technical evaluation results and aligned regional spectrum positions.

UN Involvement, Resolutions, Treaties and Events

- Resolution 253 (WRC-23), 15 DECEMBER 2023, firstly adopted at the World Radiocommunication Conference in Dubai, it encourages especially ITU to investigate mobile and satellite service allocations from quick and direct space to network connectivity. Successive to WRC-27 Agenda Item 1.13, it additionally fills the gap in ground network coverage. In addition, it instructs ITU-R to conduct spectrum sharing and frequency band study ranging 7-15 GHz Mid-band and satellite connectivity while avoiding harmful interferences.

- Recommendation ITU-R M.2160, NOVEMBER 13 2023

  The ITU Radiocommunication Assembly officially adopted these recommendations to provide guidelines on the framework and overall objectives of the future development of IMT-2030. It also demonstrates the fundamental global aims and technical requirements within standards for IMT-2030. There, six primary usage scenarios of IMT-2030 are outlined representatively including AI and ISAC usability. It serves as the central UN technical roadmap and guideline for future 6G spectrum allocation and global harmonization while alleviating digital divide.

- ITU Radio Regulations, DECEMBER 15 2024 contains the texts of Radio Regulations adopted by World Radiocommunication Conferences from 2023(WRC-23). It binds the international treaty governing global allocation of radio frequency spectrum and satellite orbits. It mandates nations to conduct technical sharing and feasibility studies especially for ITU-R Working Parties and regulates framework for Non-Terrestrial Networks.

- UN General Assembly Resolution 79/1 (The Pact for the Future), SEPTEMBER 22 2024

  Resolution of the pact for the future lists out several acts that nations take on to address issues including digital development and change. In Action 27, they state to seize the opportunities associated with new and emerging technologies and address the potential risks posed by their misuse. These are highly related to international peace and security in rapid technological change. In Action 29, they were encouraged to scale up the means of implementation to developing countries to strengthen their science, technology and innovation capacities. It

pursues to enable sustainable growth to bridge the science, technology and innovation gap between developed and developing countries.

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